

Project Report:

Facial Attendance Marking System

CMP513 + CMP514

- Project Title: Facial Attendance Marking System
- Team Members:

Sr. No.	Enrollment No.	Name
1	24000131	Krishna Patel
2	24000493	Maitra Prajapati
3	24000605	Vikas Patanwadiya

- Date of Submission:

Contents

1. Project Overview 2

 1.1 Project Title 3

 1.2 Project Category 3

 1.3 Abstract (150– 250 words) 3

2. Problem Definition 3

3. Motivation 3

4. Objectives 4

5. Scope of the Project 4

 5.1 Included Features 4

 5.2 Excluded Features 4

6. Literature Survey 4

7. Proposed Solution 5

8. System Architecture 5

5_Toc9717

10. Non-Functional Requirements 6

11. Methodology 6

12. Algorithms and Techniques 7

13. Technology Stack 7

14. Dataset 7

15. Expected Deliverables 8

16. Work Breakdown Structure 8

17. Timeline (8– 10 Weeks) 9

18. Innovation 9

19. References 10

1. Project Overview

1.1 Project Title

Facial Attendance Marking System

1.2 Project Category

Select the appropriate category

- ☐ Computer Graphics
- ☒ Image Processing
- ☐ Graphics + Image Processing
- ☐ GUI Application
- ☐ Other: _____
- ☐ Multimedia
- ☐ AR/VR
- ☐ 3D Graphics
- ☐ Game Development

1.3 Abstract (150– 250 words)

This project aims to develop an automated attendance system using facial recognition technology to eliminate manual processes, reduce proxy attendance, and provide real-time tracking for academic institutions. By leveraging advanced image processing algorithms, the system identifies students from a live camera feed and logs their presence instantly into a database, ensuring a touchless and efficient environment.

2. Problem Definition

The current educational infrastructure face several challenges regarding student tracking: manual attendance is time-consuming and prone to human error, and proxy attendance remains a significant issue in academic environments. Furthermore, teachers spend valuable lecture time on attendance rather than instruction, necessitating an automated, reliable alternative.

3. Motivation

The team aims to digitize and streamline the classroom management process, inspired by the need for touchless, efficient, and secure identification systems. This solution enhances accountability and provides digital records for further analysis.

4. Objectives

- Detect faces in real-time video streams.
- Identify registered students accurately using a robust recognition algorithm.
- Automatically log attendance time and status in a database.
- Generate attendance reports for faculty review.

5. Scope of the Project

Mention what will be implemented and what will be excluded

5.1 Included Features

- Real-time face detection
- Database integration
- Attendance logging and automated reporting

5.2 Excluded Features

- Cloud storage
- Mobile application interface
- Advanced emotion analysis

6. Literature Survey

Review **3– 5 existing tools or research papers.**

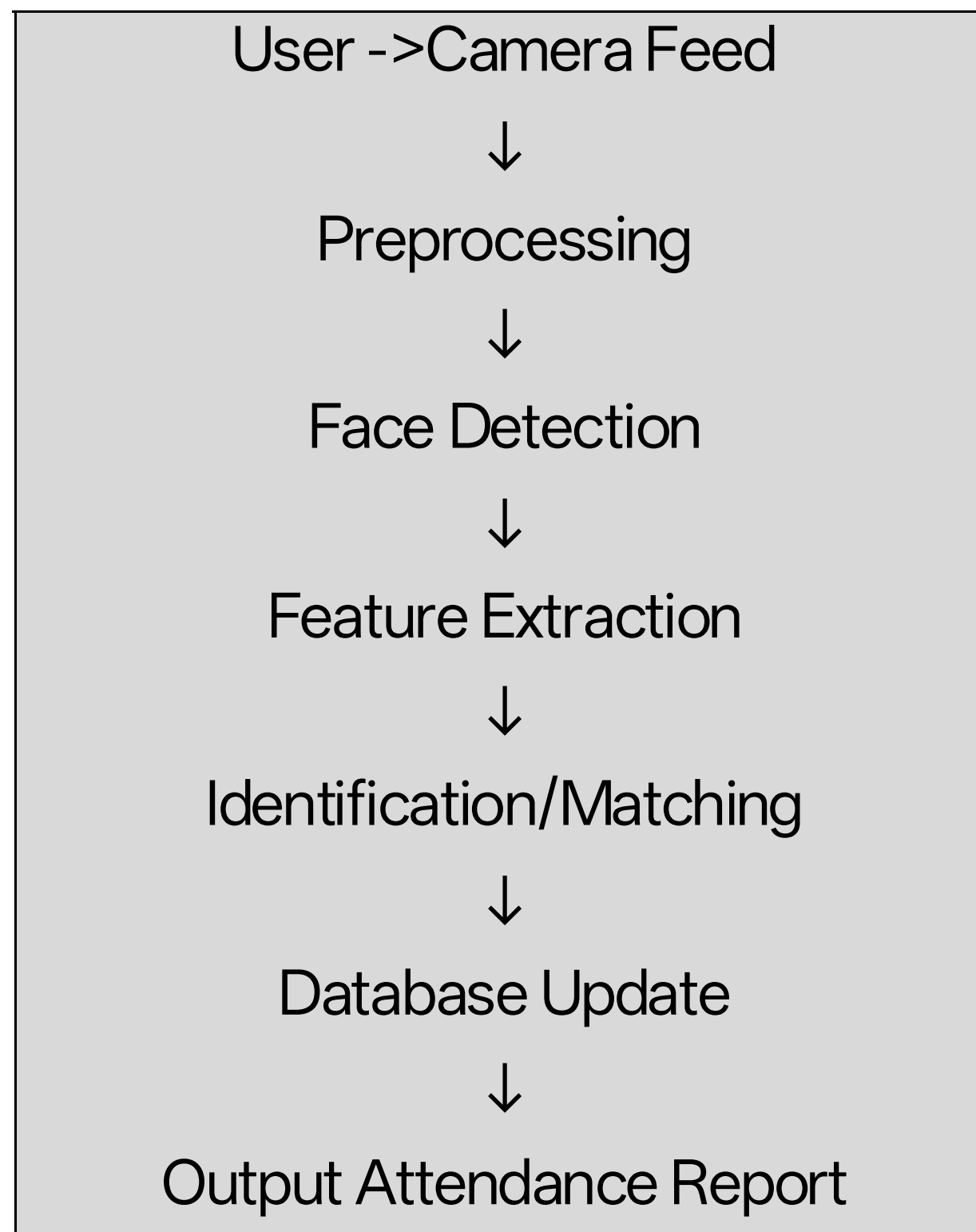
Existing Solution	Features	Limitations
Biometric Fingerprint	Accurate	Physical contact required
RFID Card System	Fast	Cards can be stolen/shared
Manual Roll Call	Simple	Time-consuming/Proxy prone

This project differs by providing a contact-less, automated, and passive attendance mechanism.

7. Proposed Solution

- Working Principle: Utilize a camera to capture a stream of frames.
- Processing Steps: Face detection -> Feature extraction -> Comparison with database -> Logging.
- Workflow: Classroom entry -> Camera detection -> Identification -> Database update.

8. System Architecture



9. Functional Requirements

1. Capture a continuous live video feed from a camera/webcam.
2. Detect one or more human faces in real time within each video frame.
3. Extract facial features and generate a unique facial embedding for each detected face.
4. Match the extracted embedding against a pre-registered student database to identify the individual.
5. Automatically mark and timestamp attendance for successfully recognized students.
6. Prevent duplicate attendance entries for the same student within a single session.
7. Provide an admin module to register new students by capturing sample face images and enrollment details.
8. Display live recognition status (Recognized / Unknown) on the GUI.
9. Generate and export attendance reports (daily/monthly) in CSV/Excel format.

10. Non-Functional Requirements

1. Real-time performance: recognition and logging completed within 1-2 seconds per face.
2. High recognition accuracy (target above 90%) under normal classroom lighting.
3. Easy-to-use, intuitive GUI for faculty with minimal training required.
4. Reliable and consistent database logging with no loss of attendance records.
5. Scalable design to accommodate a growing number of registered students.
6. Secure storage of facial data and attendance records.

11. Methodology

1. Image Acquisition: Capture live video frames from a webcam/camera using OpenCV.
2. Image Preprocessing: Convert frames to grayscale, reduce noise, and apply histogram equalization to normalize lighting variations.
3. Face Detection: Apply a face detector (Haar Cascade / HOG-based detector) to locate face bounding boxes in each frame.
4. Feature Extraction: Generate a facial embedding (e.g., 128-dimensional encoding) for each detected face using a pretrained deep learning model.
5. Face Recognition / Matching: Compare the generated embedding against the stored embeddings of registered students using distance-based matching (Euclidean distance/cosine similarity).
6. Attendance Logging: On successful match, record the student's name, enrollment number, date, and time in the attendance database, avoiding duplicate entries.
7. GUI Integration: Build a desktop interface for starting the camera feed, viewing live recognition, registering students, and exporting reports.
8. Testing: Validate the system under varying lighting conditions, angles, and partial occlusion, and compare results with manually recorded attendance.

12. Algorithms and Techniques

- Haar Cascade Classifier / HOG - for real-time face detection.
- Histogram Equalization - to normalize lighting variation across frames.
- Gaussian Filter - for noise reduction during preprocessing.
- Local Binary Patterns (LBP) - as a lightweight feature-extraction alternative.
- Deep learning embedding model (e.g., FaceNet/dlib 128-d encoder) - for robust feature extraction.
- Euclidean Distance / Cosine Similarity - for comparing facial embeddings.
- k-Nearest Neighbours (k-NN) - to classify the closest matching identity from the database.

13. Technology Stack

Component	Selected Tool
Programming Language	Python
Graphics Library	OpenCV, dlib / face_recognition library
Image Processing	TensorFlow / Keras (for embedding model)
GUI	React , Tailwind CSS ,
IDE	VS Code
Version Control	GitHub

14. Dataset :

Custom face images of enrolled students, captured using a webcam during the registration process.

15. Expected Deliverables

- Working Facial Attendance Marking System (GUI application).
- Complete source code.
- Registered face dataset/embeddings of enrolled students.
- Attendance database with exportable reports.
- Project documentation.
- Test cases and results.
- User manual.
- Presentation and demo video.

16. Work Breakdown Structure :

Below is the proposed work distribution among team members. This can be adjusted as the project progresses.

Activity	Member 1	Member 2	Member 3
Literature Survey	✓		
Face Detection Module		✓	
Feature Extraction&Recognition Module		✓	✓
Database & GUI	✓	✓	✓

Integration			
Testing & Documentation	✓		✓

17. Timeline (8– 10 Weeks)

Below table gives you the idea about the work planning

Week	Activity
1	Literature Survey
2	Requirement Analysis
3	UI Design
4	Face Detection Module Development
5	Feature Extraction&Recognition Module
6	Database Integration&Attendance Logging
7	Full System Integration
8	Testing&Bug Fixing
9	Documentation

18. Innovation

1. Fully contactless attendance mechanism, eliminating physical cards or biometric contact.
2. Real-time automated recognition that significantly reduces proxy attendance.
3. Duplicate-entry prevention logic to ensure accurate, one-time-per-session logging.
4. ghting- robust preprocessing pipeline for reliable performance across varied classroom conditions.
5. Faculty-friendly GUI with live recognition dashboard and one-click report export.

19. References

Documentation

- OpenCV Documentation - <https://docs.opencv.org>
- dlib Documentation - <http://dlib.net>

